

MAD1 PROJECT REPORT

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About me: I'm Srivalli, a Standalone student of the IIT Madras BS Degree Program. I am a passionate developer with a strong interest in web development, particularly in Flask and backend technologies..

PROJECT DETAILS

Project Title: Placement Portal Application

Problem statement: To design and build a **Placement Portal Application** web application that allows **Admin (Institute), Company, and students** to interact with the system based on their roles.

Approach:

The application was developed using Flask as the backend framework. It manages users, companies, placement drives, and student applications through structured database models.

AI/LLM Declaration

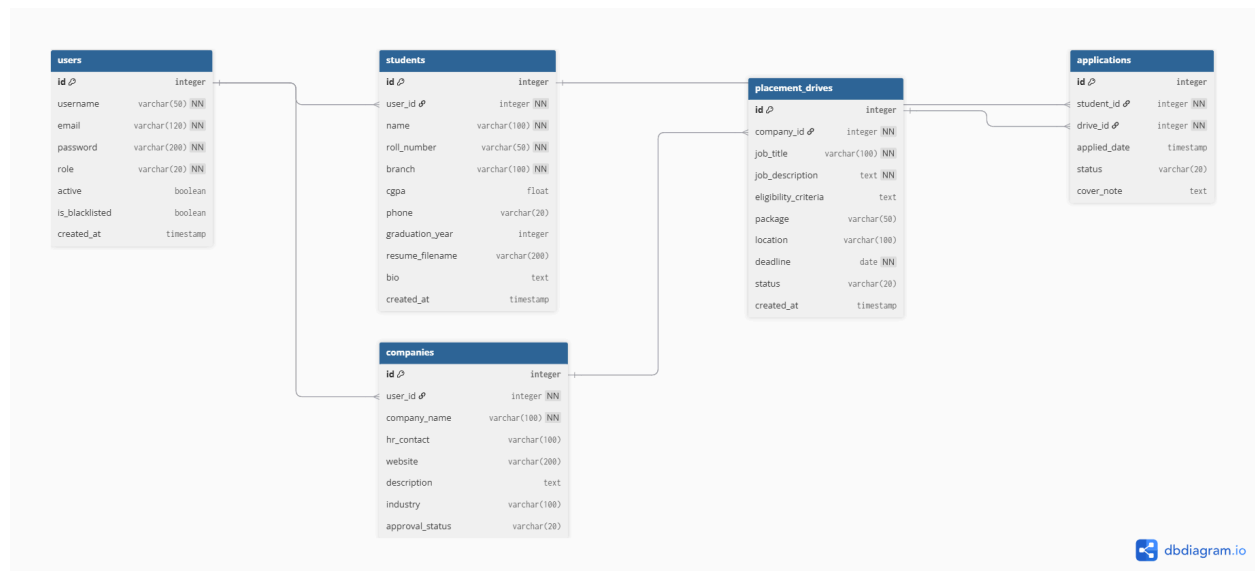
I used Claude.ai to assist with improving documentation wording, and clarifying certain backend concepts. I used ChatGPT-5 to generate basic frontend buttons and div code, which I reviewed and integrated into the project. All core implementation, database design decisions, debugging, and system integration were completed independently by me.

Technologies used :

Backend Framework	: Flask
Database	: SQLite
ORM	: SQLAlchemy
Authentication	: Flask-Login
Frontend	: HTML, CSS, Bootstrap
Templating Engine	: Jinja2

DATABASE SCHEMA / ER DIAGRAM:

USER	STUDENT	COMPANY	PLACEMENT DRIVE	APPLICATION
id (primary key)	id (primary key)	id (primary key)	id (primary key)	id (primary key)
username(unique)	user_id (foreign key)	user_id (foreign key)	company_id (foreign key)	student_id (foreign key)
Email (unique)	name	company_name	job_title	drive_id (foreign key)
password	roll_number (unique)	hr_contact	job_description	applied_date
role	branch (0-10 constraint)	website	package	status
active	phone (unique)	description	location	cover_note
is_blacklisted	graduation_year	industry	deadline	
	bio	approval_status	status	



Relationships

- One-to-One: User → Student
- One-to-One: User → Company
- One-to-Many: Company → PlacementDrive
- One-to-Many: Student → Application

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- **One-to-Many: PlacementDrive → Application**

Architecture Overview:

- **app.py** – Main Flask application entry point responsible for initializing extensions and registering blueprints.
- **/models** – Contains SQLAlchemy database models (User, Student, Company, PlacementDrive, Application).
- **/routes (Blueprints)** – Handles routing logic for admin, student, and company functionalities.
- **/templates** – Jinja2-based HTML templates for dynamic frontend rendering.
- **Authentication Layer** – Managed using Flask-Login for session handling and role-based access control.

Video Presentation:

Drive link:

https://drive.google.com/file/d/1dfiV2_Y2pSxt8XhXdJEbB7xbg2J0HV1t/view?usp=sharing